

Working with Data Types and Variables

In this lecture, we will explore data types in Python, focusing on integers, floating-point numbers, and strings. We will learn how to create and manipulate variables, perform arithmetic operations, and utilize built-in functions for data handling.

1. Introduction to Data Types

Data types are the classification of data items. Python has several built-in data types that can be used to define variables. The most commonly used data types include: - Integers: Whole numbers, e.g., 5, -3, 42 - Floating-point numbers: Decimal numbers, e.g., 3.14, -0.001, 2.0 - Strings: A sequence of characters, e.g., 'hello', "Python programming"

2. Integers

Integers are whole numbers without a fractional component. In Python, you can perform various arithmetic operations using integers. Here are some examples: - Addition: $5 + 3$ results in 8 - Subtraction: $10 - 2$ results in 8 - Multiplication: $4 * 2$ results in 8 - Division: $16 / 2$ results in 8.0 (note that this produces a floating-point number) - Modulus: $7 \% 3$ results in 1 (the remainder of the division)

3. Floating-Point Numbers

Floating-point numbers are used to represent real numbers that require a decimal point. In Python, you can perform the same arithmetic operations on floating-point numbers as you do with integers. Here are some examples: - Addition: $2.5 + 1.5$ results in 4.0 - Subtraction: $5.0 - 2.0$ results in 3.0 - Multiplication: $3.0 * 2.0$ results in 6.0 - Division: $5.0 / 2.0$ results in 2.5

4. Strings

Strings in Python are sequences of characters enclosed in single or double quotes. You can manipulate strings using various operations and methods. Here are some examples: - Concatenation: You can combine strings using the + operator. For example, "Hello" + "World" results in "Hello World". - Accessing Characters: You can access individual characters in a string using indexing. For example, in `name = 'Python'`, `name[0]` returns 'P'. - Slicing: You can extract a substring using slicing. For example, `name[1:4]` returns 'yth'. - String Length: Use the `len()` function to find the length of a string. For example, `len(name)` returns 6.

5. Variables

A variable is a named piece of memory that stores a value. In Python, variables are created using assignment statements. The syntax is as follows:

```
variable_name = value
```

For example: `x = 5` - `gpa = 3.14`

You can use variables in expressions. For example, if `x = 7`, then `x + 3` results in 10.

6. Performing Arithmetic Operations

You can perform arithmetic operations using variables. Here are some examples: - If `a = 10` and `b = 5`, then: - Addition: `result = a + b` results in 15 - Subtraction: `result = a - b` results in 5 - Multiplication: `result = a * b` results in 50 - Division: `result = a / b` results in 2.0 - Modulus: `result = a % b` results in 0

7. Built-in Functions for Data Handling

Python provides several built-in functions for data handling. Some useful functions include: - `abs(value)`: Returns the absolute value of a number. - `round(value)`: Rounds a floating-point number to the nearest integer. - `int(x)`: Converts a floating-point number to an integer. - `float(x)`: Converts an integer to a floating-point number.

8. Conclusion

In this lecture, we covered the basic data types in Python: integers, floating-point numbers, and strings. We learned how to create and manipulate variables, perform arithmetic operations, and utilize built-in functions for data handling. Understanding these concepts is crucial for writing effective Python programs.

Quiz Questions

Q1: What type of number is represented by the value 3.14 in Python?

- Integer
- Floating-point number
- String
- Boolean

Correct: Floating-point number

Q2: Which operator is used for concatenating strings in Python?

- +
- -
- *
- /

Correct: +

Q3: What function would you use to find the length of a string in Python?

- len()
- length()
- size()
- count()

Correct: len()